

Research Assignment

Section 1 Data Fundamentals

- 1) An organised place where information can be stored
used in places like banks for customer accounts
- 2) Stores data from different sources mainly for reporting and analysis
used to study historical data and help
- 3) Stores large amounts of raw data in different formats
Businesses may choose a data lake when they need flexible storage
- 4) A smaller section of data focused on one business area
usually taken from a data warehouse and mainly used by specific department
- 5) has a fixed format, has some organisations, has no fixed format
- 6) Planning how data will be organized and how different pieces of data connect
Two types conceptual and logical models; one gives basic overview other more detail
- 7) finding usually patterns and information in large amounts of data
Bank can use to detect unusual transactions
- 8) used to create, store, organize and manage databases
- 9) collecting, changing and put into another system
is important because it prepares messy data so it can be properly analysed
- 10) choose the format depending on the type and size of data working with CSV

Section B Artificial Intelligence and Machine Learning

11) Technology that allows computers to perform tasks that normally require human intelligence

Narrow and General AI one performs specific and other many tasks

12) Allows computers to learn patterns from data instead of given every rule directly
3 types supervised, unsupervised and reinforcement learning

13) type of machine learning that uses lay of neural networks to learn complex pattern
neural networks process information through connected layers and improve

14) helps computers understand and work with human language

15) AI made trained on huge amounts of text so it can understand language

16) creates new content from what it has learned

17) information used to teach on AI

If data is incorrect or bias AI can learn wrong patterns and unfair results

18) Allows computers to understand information from images and videos

Section C - APIs, tools and develop concepts

19) Allows different software programs to communicate with each other
like a waiter who takes request to kitchen and brings result

20) identifies application or user when accessing an API
should be private in the wrong hands its dangerous

21) common way of applications to communicate over the web
Get retrieves data, Post creates, Put updates and delete removes

22) Format for exchanging data

23) writing clear instructions so AI tools understand what you want

24) Applies same operation to every item in a list

25) using computer services such as software through internet
IaaS infrastructure, PaaS development SaaS ready to use software

26) keeps track changes made to files so earlier versions can be recovered
in Git commit saves changes and branch allows work separately from main

27) Interactive environment where you can write code

28) programming language popular in data analytics

Section D - Data ethics, Governance and careers

29) making personal information is collected used responsibly
European protection law

30) using data respectful way

31) Rules and process used to manage data properly with organization

32) when system produces unfair results

33) studies data, predictive models, manages data system, builds AI solution

34) uses data and visual reports to help business make better decision

Section E Personal Reflection

- 35) I found Generative AI most because I'm surprised by how much AI can create from simple instructions
- 36) API and data lakes
- 37) It will be important because we are moving in that direction due to evolving technology

Sources/References

IBM

Google Cloud

Google developers